



KTU NOTES

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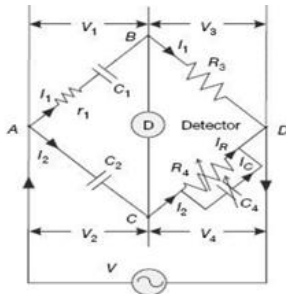
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Part A

1. Derive the equation for capacitance using Schering bridge. [3 marks] (December 2021)

&

Explain any one method to measure the leakage resistance of capacitor. [3 marks] (December 2020)



The unknown capacitor C_1 along with its internal resistance r_1 (representing loss) placed on the arm AB is compared with the standard loss-less capacitor C_2 placed on the arm AC . This capacitor C_2 is either an air or a gas capacitor to make it loss free. R_3 is a non-inductive resistance placed on arm BD . The bridge is balanced by varying the capacitor C_4 and the non-inductive resistor R_4 parallel with C_4 , placed on arm CD .

Under balanced condition, since no current flows through the detector, nodes B and C are at the same potential, i.e., $V_1 = V_2$ and $V_3 = V_4$.

Impedance of arm $AB = r_1 + 1/j\omega C_1$

Impedance of arm $CD =$ parallel combination of R_4 and $C_4 = R_4 * (1/j\omega C_4) / (R_4 + 1/j\omega C_4)$

$$= \left(\frac{R_4}{j\omega C_4} \right) * (R_4 j\omega C_4 + 1) / j\omega C_4 = R_4 / (R_4 j\omega C_4 + 1)$$

$$\text{At balance, } \frac{\left(r_1 + \frac{1}{j\omega C_1} \right)}{R_3} = \frac{\left(\frac{1}{j\omega C_2} \right)}{\left(\frac{R_4}{1 + j\omega C_4 R_4} \right)}$$

$$\text{or, } R_4 \left(r_1 + \frac{1}{j\omega C_1} \right) = \left(\frac{R_3}{j\omega C_2} \right) (1 + j\omega C_4 R_4)$$

$$\text{or, } R_4 r_1 - \frac{jR_4}{\omega C_1} = \frac{R_3 R_4 C_4}{C_2} - \frac{jR_3}{\omega C_2}$$

Equating real and imaginary parts, we have the unknown quantities:

$$r_1 = \frac{R_3 C_4}{C_2}$$

and

$$C_1 = C_2 \frac{R_4}{R_3}$$

Dissipation Factor

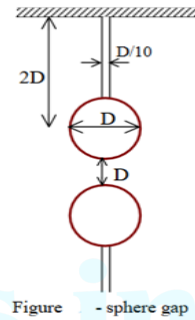
$$D_1 = \tan \delta_1 = \frac{I_1 r_1}{\frac{I_1}{\omega C_1}} = \omega C_1 r_1 = \omega \times C_2 \frac{R_4}{R_3} \times \frac{R_3 C_4}{C_2} = \omega R_4 C_4$$

2. How high voltage is tested using sphere gaps? [3 marks] (December 2021)

In the measuring device, two metal spheres are used, separated by a gas-gap. The potential difference between the spheres is raised until a spark passes between them. The breakdown strength of a gas depends on the size of the spheres, their distance apart and a number of other factors.

WORKING

- A uniform field sphere gap will always have a sparkover voltage within a known tolerance under constant atmospheric conditions.
- Hence a spark gap can be used for measurement of the peak value of the voltage, if the gap distance is known.
- A sparkover voltage of 30 kV (peak) at 1 cm spacing in air at 20°C and 760 ton pressure occurs for a Sphere Gap Measurement or any uniform field gap.
- Sphere Gap Measurement can be arranged either (i) vertically with lower sphere grounded, or (ii) horizontally with both spheres connected to the source voltage or one sphere grounded.



3. Construct a bridge circuit to measure the frequency in audio and high frequency oscillators.

[3 marks] (December 2020)

Wien's bridge is primarily used for determination of an unknown frequency

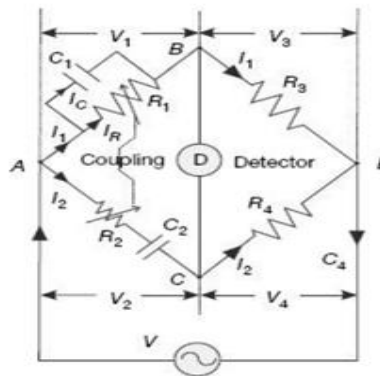


Figure **Wien's bridge under balanced condition:**

Under balanced condition, since no current flows through the detector, nodes B and C are at the same potential, i.e., $V_1 = V_2$ and $V_3 = V_4$.

At balance

$$\frac{\text{Impedance of branch AB}}{\text{Impedance of branch BD}} = \frac{\text{Impedance of branch AC}}{\text{Impedance of branch CD}}$$

Impedance of branch AB = Effective impedance of C_1 and R_1 in parallel

$$= \frac{R_1 * (1/j\omega C_1)}{R_1 + (1/j\omega C_1)} = \left(\frac{R_1}{1 + j\omega C_1 R_1} \right)$$

Similarly impedance of branch

$$AC = R_2 + 1/(j\omega C_2) = R_2 - j/\omega C_2$$

$$\text{At balance, } \frac{\left(\frac{R_1}{1 + j\omega C_1 R_1} \right)}{R_3} = \frac{\left(R_2 - \frac{j}{\omega C_2} \right)}{R_4}$$

$$\text{or, } \frac{R_1 R_4}{1 + j\omega C_1 R_1} = \frac{\omega C_2 R_2 R_3 - jR_3}{\omega C_2}$$

$$\text{or, } \omega C_2 R_1 R_4 = \omega C_2 R_2 R_3 - jR_3 + j\omega^2 C_1 C_2 R_1 R_2 R_3 + \omega C_1 R_1 R_3$$

$$\text{or, } \omega(C_2 R_1 R_4) = \omega(C_2 R_2 R_3 + C_1 R_1 R_3) - j(R_3 - \omega^2 C_1 C_2 R_1 R_2 R_3)$$

Equating real and imaginary parts, we get

$$C_2 R_1 R_4 = C_2 R_2 R_3 + C_1 R_1 R_3$$

Dividing each term by $C_2 R_1 R_3$ we get $(R_4 / R_3) = (R_2 / R_1) + (C_1 / C_2)$

$$\text{or, } \frac{R_4}{R_3} = \frac{R_2}{R_1} + \frac{C_1}{C_2}$$

and

$$\omega^2 C_1 C_2 R_1 R_2 R_3 = R_3$$

$$\text{or, } \omega = \sqrt{\frac{1}{C_1 C_2 R_1 R_2}}$$

$$\text{or, } f = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$$

In most bridges, the parameters are so chosen that,

$$R_1 = R_2 = R \text{ and } C_1 = C_2 = C$$

Then, .

$$f = \frac{1}{2\pi RC}$$

Sliders for the resistors R_1 and R_2 are mechanically coupled to satisfy the criteria $R_1 = R_2$.

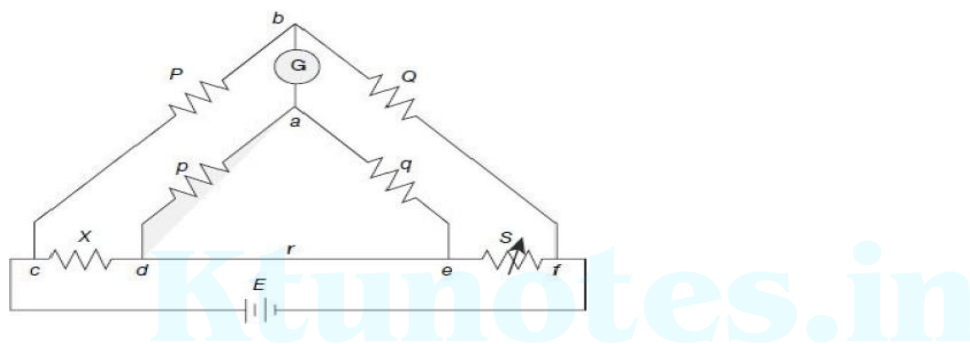
PART B

1. With the help of diagram explain the working of Kelvin double bridge. [7 marks] (December 2021)

&

Explain how low resistance is measured using kelvins Double Bridge method. Derive the balance equation used. [7 marks] (December 2020)

Kelvin's double-bridge method is one of the best available methods for measurement of low resistances.



Kelvin's double bridge incorporates the idea of the second set of ratio arms, namely, p and q , and hence the name 'double bridge'.

X is the unknown low resistance to be measured, and S is a known value standard low resistance. ' r ' is a very low resistance connecting lead used to connect the unknown resistance X to the standard resistance S . All other resistances P , Q , p , and q are of medium range. Balance in the bridge is achieved by adjusting S .

Under the balanced condition, potentials at the nodes a and b must be equal in order for the galvanometer G gives "null" deflection.

Since at balance, no current flows through the galvanometer, under balanced condition, potentials at nodes a and b are equal.

$$V_{cb} = V_{cda}$$

Now, $V_{cb} = E \times \frac{P}{P+Q}$

and $V_{cda} = V_{cd} + V_{da} = X \times I + p \times I_2$

where, $I_2 = I \times \frac{r}{r+p+q}$

$$\therefore V_{cda} = I \times X + I \times \frac{pr}{r+p+q} = I \left(X + \frac{pr}{r+p+q} \right)$$

Supply voltage $E = V_{cd} + V_{de} + V_{ef} = I \times X + I \times \frac{(p+q)}{p+q+r} \times r + I \times S$

or, $E = I \left(X + S + \frac{(p+q)}{p+q+r} \times r \right)$

we have $V_{cb} = E \times \frac{P}{P+Q}$ Substituting for E

$$V_{cb} = \frac{P}{P+Q} \times I \left(X + S + \frac{(p+q)}{p+q+r} \times r \right)$$

the balance equation $V_{cb} = V_{cda}$ can now be re-written as

$$\frac{P}{P+Q} \times I \left(X + S + \frac{(p+q)}{p+q+r} \times r \right) = I \left(X + \frac{pr}{r+p+q} \right)$$

or, $\left(X + S + \frac{(p+q)}{p+q+r} \times r \right) = \left(1 + \frac{Q}{P} \right) \times \left(X + \frac{pr}{r+p+q} \right)$

or, $X + S + \frac{(p+q)}{p+q+r} \times r = X + \frac{pr}{r+p+q} + \frac{Q}{P} \times X + \frac{Q}{P} \times \frac{pr}{r+p+q}$

or, $S + \frac{(p+q)}{p+q+r} \times r = \frac{pr}{r+p+q} + \frac{Q}{P} \times X + \frac{Q}{P} \times \frac{pr}{r+p+q}$

or, $\frac{Q}{P} \times X = S + \frac{(p+q)}{p+q+r} \times r - \frac{pr}{r+p+q} - \frac{Q}{P} \times \frac{pr}{r+p+q}$

or, $\frac{Q}{P} \times X = S + \frac{pr}{p+q+r} + \frac{qr}{p+q+r} - \frac{pr}{r+p+q} - \frac{Q}{P} \times \frac{pr}{r+p+q}$

or, $\frac{Q}{P} \times X = S + \frac{qr}{p+q+r} - \frac{Q}{P} \times \frac{pr}{r+p+q}$

or, $\frac{Q}{P} \times X = S + \frac{qr}{p+q+r} \left(1 - \frac{Q}{P} \times \frac{p}{q} \right)$

or, $X = \frac{P}{Q} \times S + \frac{qr}{p+q+r} \left(\frac{P}{Q} - \frac{p}{q} \right)$

The second quantity of the Eq. $\frac{qr}{p+q+r} \left(\frac{P}{Q} - \frac{p}{q} \right)$, can be made very small by making the ratio P/Q as close as possible to p/q . In that case, there is no effect of the connecting lead resistance 'r' on the expression for the unknown resistance. Thus, the expression for the unknown resistance X can now be simply written as

$$\text{or, } X = \frac{P}{Q} \times S$$

2. Draw the circuit and phasor diagram of Maxwell's Inductance bridge and derive the expression for unknown inductance.
 [7 marks] (December 2021)

Maxwell's inductance bridge is used to measure the value of an unknown inductance by comparing it with a variable standard self-inductance.

The bridge configuration and phasor diagram under balanced condition are shown in Figure (a) below.

The unknown inductor L_1 of resistance R_1 in the branch AB is compared with the standard known inductor L_2 of resistance R_2 on arm AC .

The inductor L_2 is of the same order as the unknown inductor L_1 .

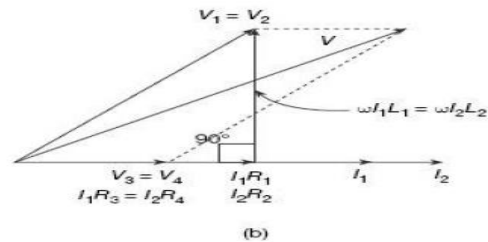
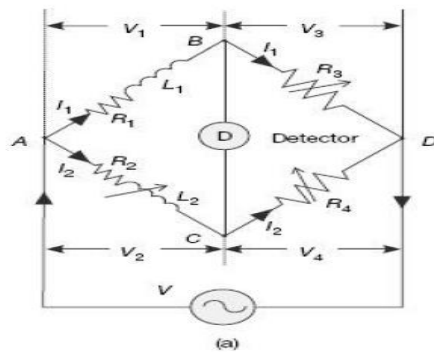
The resistances R_1, R_2 , etc., include, of course the resistances of contacts and leads in various arms.

Branch BD and CD contain known non inductive resistors R_3 and R_4 respectively.

The bridge is balanced by varying L_2 and one of the resistors R_3 or R_4 . Alternatively, R_3 and R_4 can be kept constant, and the resistance of one of the other two arms can be varied by connecting an additional resistor.

Under balanced condition, no current flows through the detector. Under such condition, currents in the arms AB and BD are equal (I_1). Similarly, currents in the arms AC and CD are equal (I_2).

Under balanced condition, since nodes B and D are at the same potential, voltage drops across arm BD and CD are equal ($V_3 = V_4$); similarly, voltage drop across arms AB and AC are equal ($V_1 = V_2$).



As shown in the phasor diagram of Figure (b), V_3 and V_4 being equal, they are overlapping.

Arms BD and CD being purely resistive, currents through these arms will be in the same phase with the voltage drops across these two respective branches.

Thus, currents I_1 and I_2 will be collinear with the phasors V_3 and V_4 .

The same current I_1 flows through branch AB as well, thus the voltage drop $I_1 R_1$ remains in the same phase as I_1 .

Voltage drop $\omega I_1 L_1$ in the inductor L_1 will be 90° out of phase with $I_1 R_1$ as shown in Figure (b). Phasor summation of these two voltage drops $I_1 R_1$ and $\omega I_1 L_1$ will give the voltage drop V_1 across the arm AB .

At balance condition, since voltage across the two branches AB and AC are equal, thus the two voltage drops V_1 and V_2 are equal and are in the same phase.

Finally, phasor summation of V_1 and V_3 (or V_2 and V_4) results in the supply voltage V .

$$\begin{aligned}\text{At balance, } \quad & \frac{R_1 + j\omega L_1}{R_3} = \frac{R_2 + j\omega L_2}{R_4} \\ \text{or, } \quad & R_1 R_4 + j\omega L_1 R_4 = R_2 R_3 + j\omega L_2 R_3 \\ \text{Equating real and imaginary parts, we have} \\ & R_1 R_4 = R_2 R_3 \\ \text{or, } \quad & \frac{R_1}{R_2} = \frac{R_3}{R_4} \\ \text{and also, } \quad & j\omega L_1 R_4 = j\omega L_2 R_3 \\ \text{or, } \quad & \frac{L_1}{L_2} = \frac{R_3}{R_4} \\ \text{Thus, } \quad & \frac{R_1}{R_2} = \frac{R_3}{R_4} = \frac{L_1}{L_2}\end{aligned}$$

Unknown quantities can hence be calculated as

$$L_1 = L_2 \times \frac{R_3}{R_4} \text{ and } R_1 = R_2 \times \frac{R_3}{R_4}$$

Care must be taken that the inductors L_1 and L_2 must be placed at a distance from each other to avoid effects of mutual inductance.

3. Explain the working of a Meggar with the help of a neat diagram. [6 marks] (December 2021)

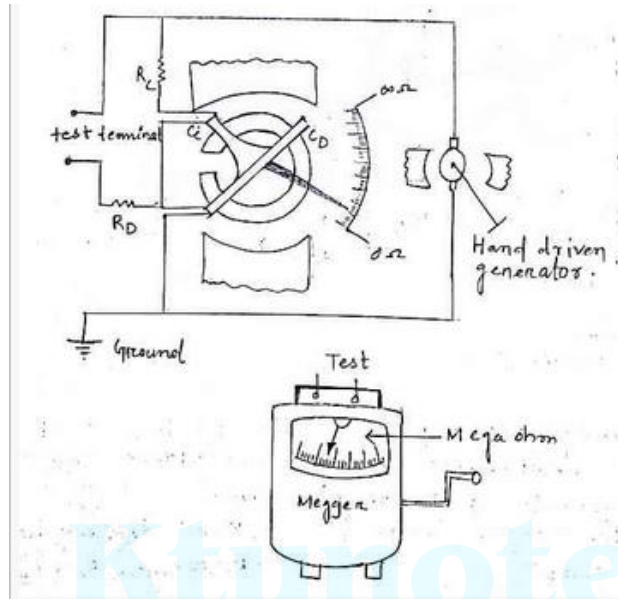
A megger is a portable device which is used for the measurement of high resistance. The component of megger are

1. Permanent magnet
2. C shaft
3. Hand driven generator

4. Resistances R_C and R_D

Two Coils-

1. Control coil (C_s)
2. Deflection Coil (C_d)



Working Principle-

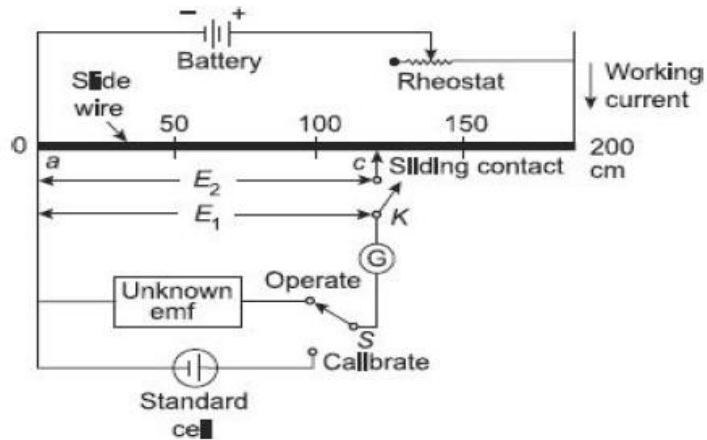
As current provided by the hand-driven generator flows through coil C , the coil will tend to set itself at a right angle to the magnetic field. The wire on coil C is wound in such a way that the torque generated is counterclockwise. Coil D is wound with the help of wire in such a way that the torque is generated in clockwise.

Initially, the test terminal is open, giving a very high value of resistance by pointing to the infinite point. In this condition there will not be any current flow in coil D . Thereby coil C will allow the motion of the pointer. When the test terminal is short-circuited the current flow through coil D and generates clockwise torque and the pointer of the ohm meter rest at 0Ω .

When a resistance with an unknown value is connected between the test terminal, current from the hand generator flows through both coils. In this situation torque in coil C and torque in coil D comes in equilibrium, and a particular value of resistance can be seen over the dial. So in this manner, we can find out the value of unknown resistance.

4. What is a DC potentiometer? Explain the calibration of ammeter and voltmeter using it with neat diagrams.

A potentiometer is an instrument which is used for measurement of potential difference across a known resistance or between two terminals of a circuit or network of known characteristics
 [8 marks] (December 2021)



A basic potentiometer circuit

Operation

First, the switch S is put in the 'operate' position and the galvanometer key K kept open, the battery supplies the working current through the rheostat and the slide wire.

The working current through the slide wire may be varied by changing the rheostat setting.

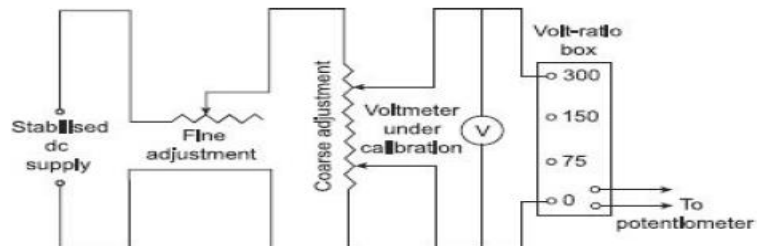
The method of measuring the unknown voltage, say E_1 , depends upon the finding a position for the sliding contact such that the galvanometer shows zero deflection, i.e., indicates null condition, when the galvanometer key K is closed.

Zero galvanometer deflection means that the unknown voltage (E_1) is equal to the voltage drop say E_2 , across position a – c of the slide wire.

When the switch S is placed at 'calibrate' position, a standard or reference cell is connected to the circuit.

This reference cell is used to standardize the potentiometer. The slide wire has a uniform cross-section and hence uniform resistance along its entire length.

Calibration of Voltmeter by Potentiometer



The arrangement for calibrating a voltmeter by potentiometer is shown in Figure.

The potential divider network consists of two rheostats. One for coarse and the other for fine control of calibrating voltage. With the help of these controls, it is possible to adjust the supply voltage so that the pointer coincides exactly with a major division of the voltmeter.

The voltage across the voltmeter is stepped down to a value suitable for the potentiometer with the help of the volt-ratio box. In order to get accurate measurements, it is necessary to measure voltages near the maximum range of the potentiometer, as far as possible.

The potentiometer measures the true value of the voltage.

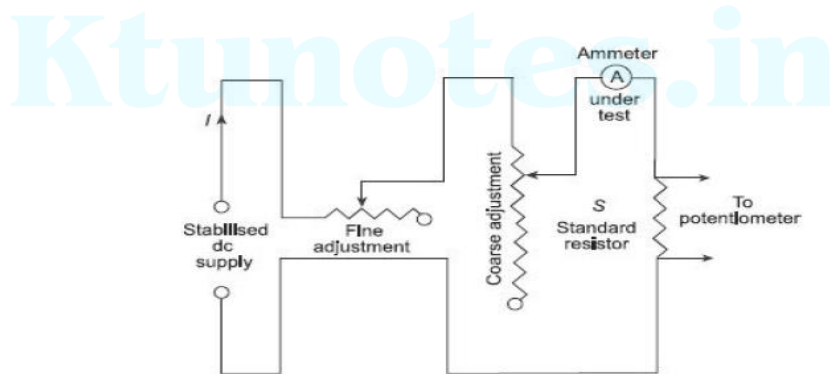
If the reading of the potentiometer does not match with the voltmeter reading, a positive or negative error is indicated.

A calibration curve may be drawn with the help of the potentiometer and the voltmeter reading.

Calibration of Ammeter by Potentiometer

Figure below shows the circuit arrangement for calibration of an ammeter using potentiometer.

A standard resistor S of high current carrying capacity is placed in series with the ammeter under test. The voltage drop across S measured with the help of the potentiometer and then the current through S



Calibration of ammeter by potentiometer

and hence the ammeter can be computed by dividing the voltage drop by the value of the standard resistor.

Current, $I = V_s / S$ is the voltage drop across the standard resistor S .

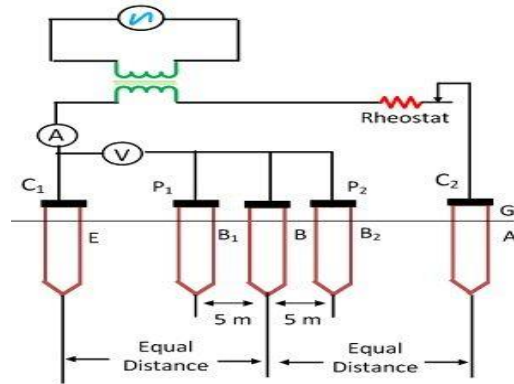
Now, compare the reading of the ammeter with the current found by calculation. If they do not match, a positive or negative error will be induced.

A calibration curve may be drawn between the ammeter reading and the true value of the current as indicated by the potentiometer reading.

As the resistance of the standard resistor S is exactly known, the current through S is exactly calculated. This method of calibration of ammeter is very accurate.

Explain any one method used for the measurement of earth resistance. What are the factors on which the resistance of earthing system depends?. [7 marks] (December 2020)

The measurement of earth resistance is made by the **potential fall method**. The resistance area of the earth electrode is the area of the soil around which a voltage gradient is measured with a commercial instrument. In the figure shown below E is the earth electrode under test, and A is an auxiliary earth electrode positioned so that two resistance areas do not overlap. B is the second auxiliary electrode which is placed between E and A.



An alternating current of steady state value passes through the earth path from E to A and the voltage drop between E and B is measured.

$$R_e = \frac{\text{Voltage drop between E and B}}{\text{Current through earth path}}$$

$$R_e = \frac{V}{I}$$

The electrode B is moved from position B₁ and B₂ respectively so that the resistance area do not overlap. If the resistance values determined are of approximately the same in all three cases, the mean of the three readings can be taken as the earth resistance of the earth electrode.

The auxiliary earth electrode A must be driven in at a point further away from E and the above test repeated until the group of three readings obtained are in good agreement.

The alternating current source is used to eliminate the electrolytic effect.

Some of the factors that affect an earthing installation are as follows:

- **Soil Condition**

The condition of soil plays a key role in determining the efficacy of earthing. The resistance of the earth, the salt, and the moisture level of the soil tell earthing specialists as to how an earthing is made. For example, earthing in rocky soil is different from earthing in wet soil.

- **Soil Resistivity**

Soil resistivity is another factor that affects earthing installations. Different conditions of soil provide different kinds of soil resistivity. A majority of soil conditions are not adaptable to earthing installations.

- **Dissolved Salts**

Pure water is considered a bad conductor of electricity. Soil resistivity depends on water resistivity which then depends on the quantity as well as nature of salts that are dissolved in it.

- **Moisture in Soil**

Moisture that exists in the soil is tested by considering the amount of water contained by the soil & its water resistivity. Hence, it is very important to provide water in the earth and around it for maintaining moisture in a dry climate.

- **Location of Earth Pit**

The location where the earthing is done is also of great value.

- **Physical Composition**

A variety of soil compositions offers different resistivity. Clay soil has a 4-150 ohm-meter resistivity range, whereas the range for gravel or rocky soils may be above than 1000 ohm-meter.

- **Climate Condition**

The climate condition also affects the earthing process. When the moisture in the soil increases or decreases, the soil is increased or decreased respectively. So, in a dry climate, the resistivity is high & in monsoons, the resistivity becomes low.

Explain how relative permittivity can be measured using Schering Bridge. [4 marks] (December 2020)

&

With neat circuit diagram and phasor explain how capacitance is measured using Schering Bridge. What is dissipation factor and derive its equation. [10 marks] (December 2020)

(For 4 marks question, only the diagram and relevant equations need to be written. Then write how to calculate ϵ_r from C. Capacitance is given by the expression $C = \epsilon_0 \epsilon_r A/t$. The value of C can be measured using schering bridge. Knowing Area A, thickness t of the dielectric, and ϵ_0 , ϵ_r can be calculated.)

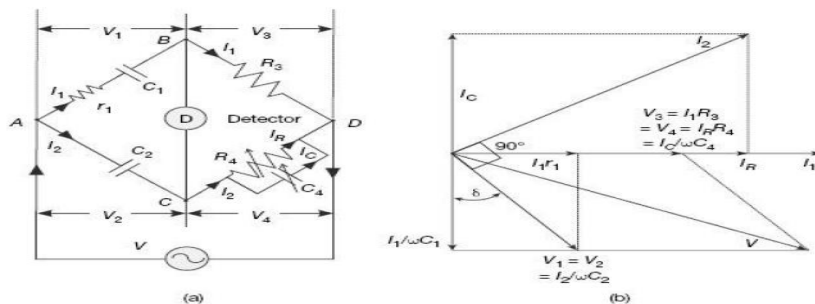


Figure — Schering bridge under balanced condition: (a) Configuration (b) Phasor diagram

EET 203 Measurements & Instrumentation
Module III: University Questions and Answers

Prepared by Dr. Renuka T K , EEE, MESCE

The unknown capacitor C_1 along with its internal resistance r_1 (representing loss) placed on the arm AB is compared with the standard loss-less capacitor C_2 placed on the arm AC . This capacitor C_2 is either an air or a gas capacitor to make it loss free. R_3 is a non-inductive resistance placed on arm BD . The bridge is balanced by varying the capacitor C_4 and the non-inductive resistor R_4 parallel with C_4 , placed on arm CD .

Under balanced condition, since no current flows through the detector, nodes B and C are at the same potential, i.e., $V_1 = V_2$ and $V_3 = V_4$.

Impedance of arm $AB = r_1 + 1/j\omega C_1$

Impedance of arm $CD =$ parallel combination of R_4 and $C_4 = R_4 * (1/j\omega C_4) / (R_4 + 1/j\omega C_4)$

$$= \left(\frac{R_4}{j\omega C_4} \right) * (R_4 j\omega C_4 + 1) / j\omega C_4 = R_4 / (R_4 j\omega C_4 + 1)$$

$$\text{At balance, } \frac{\left(r_1 + \frac{1}{j\omega C_1} \right)}{R_3} = \frac{\left(\frac{1}{j\omega C_2} \right)}{\left(\frac{R_4}{1 + j\omega C_4 R_4} \right)}$$

$$\text{or, } R_4 \left(r_1 + \frac{1}{j\omega C_1} \right) = \left(\frac{R_3}{j\omega C_2} \right) (1 + j\omega C_4 R_4)$$

$$\text{or, } R_4 r_1 - \frac{jR_4}{\omega C_1} = \frac{R_3 R_4 C_4}{C_2} - \frac{jR_3}{\omega C_2}$$

Equating real and imaginary parts, we have the unknown quantities:

$$r_1 = \frac{R_3 C_4}{C_2}$$

and

$$C_1 = C_2 \frac{R_4}{R_3}$$

Dissipation Factor

$$D_1 = \tan \delta_1 = \frac{I_1 r_1}{\frac{I_1}{\omega C_1}} = \omega C_1 r_1 = \omega \times C_2 \frac{R_4}{R_3} \times \frac{R_3 C_4}{C_2} = \omega R_4 C_4$$

Capacitance is given by the expression $C = \epsilon_0 \epsilon_r A/t$. The value of C can be measured using the Schering bridge. Knowing Area A , thickness t of the dielectric, and ϵ_0 , ϵ_r can be calculated